



September 2016 Examination

EC400

Probability and Statistical Inference

Instructions to candidates

Time allowed: 2 hours

This paper contains **SEVEN** questions. Answer **ALL** questions.

You are not allowed to use an electronic calculator

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1. **(8 points - 10 minutes)** Let us take a fair coin and flip it 3 times. What is the probability of getting a tail on the third flip given that

(a) There is a tail on the first flip?

2 points: *By independence $1/2$*

(b) There is a tail on the the first two flips?

2 points: *By independence $1/2$*

(c) There is a total of one tail?

2 points: *This is asking for $\Pr(X_3 = 1 | X_1 + X_2 + X_3 = 1)$* $\stackrel{\text{Bayes' Thm}}{=} \frac{\Pr(X_3=1 \cap X_1+X_2+X_3=1)}{\Pr(X_1+X_2+X_3=1)} = \frac{\Pr(X_1=0, X_2=0, X_3=1)}{\binom{3}{1}(1/2)^3} = \frac{1}{3}$

(d) There is a total of two tails?

2 points: *This is asking for $\Pr(X_3 = 1 | X_1 + X_2 + X_3 = 2)$* $\stackrel{\text{Bayes' Thm}}{=} \frac{\Pr(X_3=1 \cap X_1+X_2+X_3=2)}{\Pr(X_1+X_2+X_3=2)} = \frac{\Pr(X_1=0, X_2=1, X_3=1) + \Pr(X_1=1, X_2=0, X_3=1)}{\binom{3}{2}(1/2)^3} = \frac{2}{3}$

Maximum points awarded 6 if no mention is made of Bayes' Theorem.

2. **(8 points - 10 minutes)** Let X and Y be two random variables and $Y = -\sqrt{X+1}$. The probability density function of X is:

$$f_X(x) = \frac{1}{9}x^2, 0 \leq x \leq 3$$

$$= 0 \text{ elsewhere.}$$

What is the probability density function of Y ?

We observe that $Y = u(X)$ with $u(X) = -\sqrt{X+1}$ is a one-to-one transformation whose inverse is given by $v(Y) = Y^2 - 1$, and $v'(Y) = 2Y$. Using the change of variable formula

$$g_Y(y) = f_X(Y^2 - 1) \cdot |2y| \text{ with } -2 \leq y \leq -1$$

$$= \frac{1}{9} (y^2 - 1)^2 \cdot (-2y)$$

$$= -\frac{2}{9} y (y^2 - 1)^2, \quad -2 \leq y \leq -1$$

3 points: *use of proper formula to obtain marginal of Y*

3 points: *proper execution of formula*

1 point: *statement that formula uses the one-to-one relation between the r.v.'s*

1 point: *to properly state $-2 \leq y \leq -1$*

3. (12 points - 15 minutes) Let (X, Y) have a joint density function

$$f_{X,Y}(x, y) = \begin{cases} 2 & ; \quad 0 < x < y < 1 \\ 0 & ; \quad \text{otherwise} \end{cases}.$$

Derive the unconditional and conditional expectation of Y ($E(Y)$ and $E(Y|X = x)$) and discuss the relation between them.

Answer: $f(x) = \int_x^1 2dy = 2(1-x)$, $0 < x < 1$

$f(y|x) = \frac{f(y,x)}{f(x)} = \frac{2}{2(1-x)} = \frac{1}{1-x}$, $x < y < 1$ with $0 < x < 1$ and

$E(Y|X = x) = \int_x^1 y \frac{1}{1-x} dy = \frac{1+x}{2}$, $0 < x < 1$

By law of iterated expectations: $E(Y) = E_X[E(Y|X = x)] = \int_0^1 \frac{1+x}{2} f(x) dy = \int_0^1 (1+x)(1-x) dy = x - \frac{1}{3}x^3 \Big|_0^1 = \frac{2}{3}$

Student could also provide $E(Y)$ by first deriving $f(y)$, but student is expected to relate both expectation by means of law of iterated expectations.

4 points: proper definitions of conditional and unconditional expectations

4 points: proper execution of formula

4 points: law of iterated expectations

4. (12 points - 15 minutes) Let X and Y be two random variables. Show:

(X, Y) independent $\Rightarrow Y$ mean independent of $X \Rightarrow X, Y$ uncorrelated.

- **6 points:** Y is mean independent of X if $E(Y|X = x) = E(Y)$. By definition

$$E(Y|X = x) = \int y f(y|x) dy = \int y \frac{f(y, x)}{f_X(x)} dy$$

By independence we know $f(y, x) = f_X(x)f_Y(y)$, therefore

$$E(Y|X = x) = \int y \frac{f_X(x)f_Y(y)}{f_X(x)} dy = \int y f_Y(y) dy$$

with the latter by definition equalling $E(Y)$.

- **6 points:** X and Y are uncorrelated means $\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0$. So we need to show $E(XY) = E(X)E(Y)$ when Y is mean independent of X . By law of iterated expectations: $E(XY) = E_X(E(XY|X = x))$. As X is no longer stochastic when conditioning on X , this equals $E_X(XE(Y|X = x))$. Using the mean independence, we rewrite this as $E_X(XE(Y)) = E(Y)E_X(X) = E(Y)E(X)$ which proves that X and Y indeed are uncorrelated. **[Only give 2 points when only shown that independence implies uncorrelatedness here]**

5. **(18 points - 20 minutes)** Let the random variable X be normally distributed with mean μ and variance 1. You are given a random sample of 100 observations.

(a) Construct a 95% confidence interval for μ .

6 points. We know that $\bar{X} \sim N(\mu, \frac{1}{100})$ or $10(\bar{X} - \mu) \sim N(0, 1)$

$$\Pr(-1.96 < 10(\bar{X} - \mu) < 1.96) = 0.95$$

$$\Pr\left(\bar{X} - \frac{1.96}{10} < \mu < \bar{X} + \frac{1.96}{10}\right)$$

so that the 95% confidence interval is given by

$$\left[\bar{X} - \frac{1.96}{10}, \bar{X} + \frac{1.96}{10}\right]$$

For our sample it is given by:

$$\left[\bar{x} - \frac{1.96}{10}, \bar{x} + \frac{1.96}{10}\right]$$

Award only 3 points when only interval is given (no explanation how it is obtained). Could give 2 points if all that is provided is that $\bar{X} \sim N(\mu, \frac{1}{100})$ I suppose.

(b) Show that the rejection of the null $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$ with a 5% level of significance, corresponds to the rejection of H_0 when zero does not lie in the 95% confidence interval for μ given in part a).

12 points ..(Give 6 points for clear discussion of acceptance rule of H_0 with 5% level of significance, and add 6 to show that this rule conforms to 0 not lying in the 95% confidence boundary)

To say that 0 does not lie in the 95% confidence interval means

$$0 < \bar{x} - \frac{1.96}{10} \text{ or } 0 > \bar{x} + \frac{1.96}{10}$$

which is the same the observation that

$$\bar{x} > \frac{1.96}{10} \text{ or } \bar{x} < -\frac{1.96}{10}$$

When we conduct the test of $\mu = 0$ at the 5% level of significance, we could use the statistic $Z = \frac{\bar{X} - 0}{\sqrt{1/100}}$ which under H_0 is $N(0, 1)$. Our decision to reject in

favour of $\mu \neq 0$ is to reject if $z > 1.96$ or $z < -1.96$. Because $z = \bar{x}/10$, we realize that this rejection rule is the same as rejecting when $\bar{x} > \frac{1.96}{10}$ or $\bar{x} < -\frac{1.96}{10}$, which is the same as saying that 0 does not lie in the confidence interval.

6. **(25 points - 30 minutes)** Let $X_1, \dots, X_n$ be a random sample drawn from $N(\theta, 4)$. You want to test the hypothesis that $H_0 : \theta = 5$ against the alternative $H_1 : \theta > 5$.

(a) Discuss how you would conduct the above test against the one-sided alternative using a 5% level of significance.

4 points: You would use the Z test, where $Z = \frac{\bar{X}-5}{\sqrt{4/n}}$. Under the null $Z \sim N(0, 1)$, and we reject if our realization $z > 1.645$. (critical value need to be clearly indicated).

(b) Provide the power of this test at θ_1 where $\theta_1 > 5$. Discuss how the power is affected by the distance of θ_1 from 5 and the sample size. Interpret these findings.

3 points: $Power(\theta_1) = \Pr(\text{Reject } H_0 | \theta = \theta_1) = \Pr\left(\frac{\bar{X}-5}{\sqrt{4/n}} > 1.645 | \theta = \theta_1\right)$

$= \Pr\left(\frac{\bar{X}-\theta_1}{\sqrt{4/n}} > 1.645 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right) | \theta = \theta_1\right) = 1 - \Phi\left(1.645 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right)\right)$ because under the alternative $\frac{\bar{X}-\theta_1}{\sqrt{4/n}} \sim N(0, 1)$

3 points: The power is increasing when θ_1 becomes bigger relative to 5, ultimately yielding a power equal to 1. If the truth is far away from the null we should reject it with probability 1.

3 points: As the sample size increases, the precision of our estimator improves ($Var(\bar{X}) = 4/n$ becomes smaller.) As a result it is easier to reject the null (Z statistic becomes bigger), so also when the null is false. Thus increases the power as well.

(c) Specify how you would obtain the power at θ_1 if you had used a two-sided alternative instead with the same level of significance. In general, how does this compare with the power obtained in (b)? Discuss.

If we have used a two sided test, we need to define the power as

2 points: $Power(\theta_1) = \Pr(\text{Reject } H_0 | \theta = \theta_1) = \Pr\left(\frac{\bar{X}-5}{\sqrt{4/n}} > 1.96 \cup \frac{\bar{X}-5}{\sqrt{4/n}} < -1.96 | \theta = \theta_1\right)$

2 points: $\Pr\left[\frac{\bar{X}-\theta_1}{\sqrt{4/n}} > 1.96 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right) \cup \frac{\bar{X}-\theta_1}{\sqrt{4/n}} < -1.96 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right) | \theta = \theta_1\right] = \Phi\left(-1.96 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right)\right) + 1 - \Phi\left(1.96 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right)\right)$.

3 points: The power associated with a one-sided test typically is bigger. Note if θ_1 is much bigger than 5, $\Phi\left(-1.96 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right)\right)$ is negligible, so we only have to compare $1 - \Phi\left(1.96 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right)\right)$ with $1 - \Phi\left(1.645 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right)\right)$, and clearly

the latter is bigger... $[1.645 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right) < 1.96 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right)$, and by monotonicity of Φ : $\Phi\left(1.645 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right)\right) < \Phi\left(1.96 + \left(\frac{5-\theta_1}{\sqrt{4/n}}\right)\right)$ A graph may be useful here. Most power comes from the area where we reject, and having a lower critical value with the one-sided test, we reject earlier, so also when the null is false.

- (d) For a given sample of 100 observations, you obtain a point estimate $\bar{x} = 5.2$. Obtain the p-value associated with the one-sided test. Clearly indicate how we should interpret the p-value.

2 points: The p-value gives the lowest level of significance at which we want to reject the null.

3 points: For the one-sided test, the p-value is given by: $\Pr(\bar{X} > 5.2) = \Pr\left(\frac{\bar{X}-5}{\sqrt{4/100}} > \frac{5.2-5}{\sqrt{4/100}}\right) = 1 - \Phi(1) = 1 - .841 = .159$.

7. **(18 points - 20 minutes)** Let X is a random variable from the Pareto distribution, which has density

$$f(x) = \frac{\theta}{x^{\theta+1}}, x > 1, 1 < \theta < \infty$$

- (a) Obtain the expected value of X .

$$E(X) = \int_1^\infty x \frac{\theta}{x^{\theta+1}} dx = \int_1^\infty \theta x^{-\theta} dx = -\frac{\theta}{\theta-1} x^{-(\theta-1)} \Big|_1^\infty = \frac{\theta}{\theta-1}$$

6 points. 3 points if only definition is given (correctly) and no attempt at evaluating integral was made. "Silly" error in derivations, 1 point penalty.

- (b) A sample of n independent realizations of X ($\{x_1, \dots, x_n\}$) was collected

- i. Find the maximum likelihood estimator for θ ($\hat{\theta}_{MLE}$).

By independence the likelihood can be expressed as

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n f(x_i) \\ &= \frac{\theta^n}{x_1^{\theta+1} \dots x_n^{\theta+1}} \\ \ln L(\theta) &= n \ln(\theta) - (\theta+1) \sum_{i=1}^n \ln(x_i) \\ \text{F.O.C.: } \frac{\partial \ln L}{\partial \theta} &: \frac{n}{\hat{\theta}_{MLE}} - \sum_{i=1}^n \ln(x_i) = 0 \\ \text{or } \hat{\theta}_{MLE} &= \left(\frac{1}{n} \sum_{i=1}^n \ln(x_i) \right)^{-1} \end{aligned}$$

3 points: provide the likelihood function (subtract 1 point if use of independence is not clearly referred to)

3 points: FOC (properly using $\hat{\theta}_{MLE}$) and solve (subtract 1 point if θ is used instead because FOC only need to be satisfied for $\hat{\theta}_{MLE}$)

- ii. Find the method of moments estimator for θ ($\hat{\theta}_{MME}$)

From the expectation result in a, we choose θ in such a way that its sample analogue is satisfied, i.e.,

$$\bar{x} = \frac{\hat{\theta}_{MME}}{\hat{\theta}_{MME} - 1}$$
$$(\hat{\theta}_{MME} - 1)\bar{x} = \hat{\theta}_{MME} \text{ or } \hat{\theta}_{MME} = \frac{\bar{x}}{\bar{x} - 1}$$

3 points: if a) was incorrect but student clearly explains the idea of MME.

6 points: if correctly solves for $\hat{\theta}_{MME}$ (1 point penalty for "silly" error in derivation)